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python-programing-mechine-learning / supervised_learning_(9_1).ipynb

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93 lines (93 loc) · 50 KB

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In [ ]:

```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn import tree
from sklearn.tree import DecisionTreeClassifier
import matplotlib.pyplot as plt

df = pd.DataFrame({'Age':[36,42,23,52,43,44,66,35,52,35,24,18,45],
'Experience':[10,12,4,4,21,14,3,14,13,5,3,3,9],
'Rank':[9,4,6,4,8,5,7,9,7,9,5,7,9],
'Nationality':['UK','USA','N','USA','USA','UK','N','UK','N','N','USA','UK','UK'],
'Go':['NO','NO','NO','NO','YES','NO','YES','YES','YES','YES','NO','YES','YES']
})
print(df)

d = {'UK': 0, 'USA': 1, 'N': 2}
df['Nationality'] = df['Nationality'].map(d)
d = {'YES': 1, 'NO': 0}
df['Go'] = df['Go'].map(d)

features = ['Age', 'Experience', 'Rank', 'Nationality']
X = df[features]
y = df['Go']

dtree = DecisionTreeClassifier()

dtree = dtree.fit(X,y)

tree.plot_tree(dtree, feature_names=features)
plt.show()
```

```
    Age  Experience  Rank Nationality   Go
0    36          10     9          UK   NO
1    42          12     4         USA   NO
2    23           4     6           N   NO
3    52           4     4         USA   NO
4    43          21     8         USA  YES
5    44          14     5          UK   NO
6    66           3     7           N  YES
7    35          14     9          UK  YES
8    52          13     7           N  YES
9    35           5     9           N  YES
10   24           3     5         USA   NO
11   18           3     7          UK  YES
12   45           9     9          UK  YES
```